

Module 1 — Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 1, Quiz A: Definitions & notation

Reading: [Module 1 Read 1](#)

Canvas: Def & Notation quiz (Module 1)

Work these **practice** items after [Read 1](#). Sections follow the read order (§1–§4). Same ideas appear in **Check yourself** boxes. **No R code here** — R items are on the [Synthesis](#) bank (after the demo lab).

Section A — Probability language (Read 1 §1)

A1. Counts from data

A student flips a fair coin **20 times** and observes **9 heads**. What are x and n ?

A2. Observed proportion

A fair coin is tossed **20 times** and **9 heads** are observed, so $x = 9$ and $n = 20$. Compute $\hat{p}$. Show your calculation.

A3. Percentage

In a study, the observed sample proportion of successes is $\hat{p} = 0.45$. Convert this to a percentage.

A4. Data vs theory

A fair coin is tossed **20 times** and **9 heads** are observed ($\hat{p} = 9/20 = 0.45$). For a fair coin, the long-run probability of heads is 0.5. Using correct notation, state the **observed proportion** of heads and the **theoretical** probability of heads.

A5. $\hat{p}$ vs $P(\text{event})$

In one or two sentences, explain the difference between $\hat{p}$ and $P(\text{event})$. Refer to **data** and **long-run behavior**.

A6. Definition of probability

State the definition of probability involving a **repeatable random process**, a **proportion**, and the idea that the process continues **forever**.

Answer key — Section A — Probability language (Read 1 §1)

1. $x = 9$ (count of heads); $n = 20$ (total tosses).
 2. $\hat{p} = x/n = 9/20 = 0.45$.
 3. 45%.
 4. Observed: $\hat{p} = 0.45$. Theoretical: $P(\text{head}) = 0.5$.
 5. $\hat{p}$ summarizes **observed data** from a fixed sample. $P(\text{event})$ describes **long-run** behavior of the random process.
 6. $P(\text{Event})$ is the proportion of times the event occurs when the trial is repeated infinitely many times.
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Section B — Sample space, outcomes, and events (Read 1 §2)

B1. Vocabulary

Give a clear definition of each term:

- a) **Sample space**
- b) **Outcome**
- c) **Event**

B2. Die trial vocabulary

One roll of a fair six-sided die:

1. Write the sample space S .
2. Give one **outcome**.
3. Give one **event** (and list the outcomes in it).

B3. Set language

Suppose a trial has sample space S and events A and B .

1. In words, what does “ A and B ” mean?
2. In words, what does “ A or B ” mean?
3. In words, what does “not A ” mean?

B4. Translate notation to English

Write a one-sentence interpretation for each:

1. $P(A) = 0.7$
2. $P(A \text{ and } B) = 0.2$
3. $P(\text{not } A) = 0.1$

Answer key — Section B — Sample space, outcomes, and events (Read 1 §2)

1. a) **Sample space** S : all possible outcomes of a trial. b) **Outcome**: one element of S . c) **Event**: a set of one or more outcomes.
 2. $S = \{1, 2, 3, 4, 5, 6\}$; e.g. outcome 4; e.g. $A = \{2, 4, 6\}$ (even).
 3. **And**: both occur. **Or**: at least one occurs. **Not** A : A does not occur.
 4. 1. There is a 70% chance event A occurs. 2. There is a 20% chance both A and B occur. 3. There is a 10% chance A does not occur.
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Section C — Equally likely outcomes (Read 1 §2)

C1. Equally likely rule

State the probability rule for event A when all outcomes in S are equally likely. Define $|A|$ and $|S|$ in words.

C2. When the rule applies

State the **condition** that must be true for $P(A) = |A|/|S|$ to be valid.

Answer key — Section C — Equally likely outcomes (Read 1 §2)

1. $P(A) = |A|/|S|$ where $|A|$ = number of outcomes in A , $|S|$ = number in the sample space.
 2. All outcomes in the sample space must be **equally likely**.
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Section D — Addition rule (Read 1 §3)

D1. Addition rule

State the **addition rule** for events A and B .

D2. Backpacks and hats

65 students: 40 have backpacks, 10 have hats, 5 have **both**. Let A = has backpack, B = has hat.

1. Find $P(A)$ and $P(B)$.
2. Find $P(A \text{ and } B)$.
3. Use the addition rule to find $P(A \text{ or } B)$.

D3. Complement

If $P(\text{rains at least once in July}) = 0.91$, find $P(\text{no rain in July})$.

D4. Disjoint events

Suppose A and B are **non-overlapping** (mutually exclusive).

1. What does “non-overlapping” mean in words?
2. What is $P(A \text{ and } B)$?
3. Write the simplified formula for $P(A \text{ or } B)$.

Answer key — Section D — Addition rule (Read 1 §3)

1. $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.
 2. $P(A) = 40/65$, $P(B) = 10/65$, $P(A \text{ and } B) = 5/65$, $P(A \text{ or } B) = 45/65$.
 3. $1 - 0.91 = 0.09$.
 4. They cannot both occur; $P(A \text{ and } B) = 0$; $P(A \text{ or } B) = P(A) + P(B)$.
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Section E — Conditional probability and independence (Read 1 §4)

E1. Conditional probability

1. In words, what does $P(A | B)$ mean?
2. State the formula for $P(A | B)$ when $P(B) > 0$.

E2. Multiplication rule

1. State the **multiplication rule** for $P(A \text{ and } B)$.
2. State the **special case** when A and B are independent.

E3. Independence

1. In words, when are A and B **independent**?
2. Give **two** equivalent ways to check independence.

E4. Translate conditional notation

Write a one-sentence interpretation for each:

1. $P(A | B) = 0.3$
2. $P(A \text{ and } B) = P(A)P(B)$

E5. Match notation with English

Define A = has the disease, B = tests positive. **Match** each expression to the correct interpretation:

Expression	Interpretation
$P(A \text{ and } B)$	(a) Baseline chance of disease before any test
$P(A)$	(b) Among positive tests, chance of disease
$P(A B)$	(c) Proportion with disease and positive test
$P(B A)$	(d) Among diseased, chance test is positive
$P(A \text{ or } B)$	(e) Has disease, or tests positive, or both

E6. Die conditionals

Fair die rolled once:

1. Find $P(\text{roll is } 2 | \text{roll is even})$.
2. Find $P(\text{roll is even} | \text{roll is } 2)$.

Answer key — Section E — Conditional probability and independence (Read 1 §4)

1. Probability of A given B occurred: $P(A | B) = P(A \text{ and } B)/P(B)$.
 2. $P(A \text{ and } B) = P(A | B)P(B)$; if independent, $P(A \text{ and } B) = P(A)P(B)$.
 3. Knowing one does not change the other's probability; e.g. $P(A | B) = P(A)$ or $P(A \text{ and } B) = P(A)P(B)$.
 4. 1. Given B occurred, there is a 30% chance A occurs. 2. A and B are independent (joint equals product of marginals).
 5. $P(A \text{ and } B) \rightarrow (c)$; $P(A) \rightarrow (a)$; $P(A | B) \rightarrow (b)$; $P(B | A) \rightarrow (d)$; $P(A \text{ or } B) \rightarrow (e)$.
 6. 1. $1/3$. 2. 1 (if the roll is 2, it is certainly even).
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